Project: PORTAL GUN
Investigation: I want to create a portal gun

1. Abstract & Real-World Literature

This paper investigates the given goal by consulting current scientific literature.

Literature retrieval is disabled: ideas come from the model's own knowledge and are NOT grounded in retrieved sources.

2. Methodology & Viability Metrics

A. SCIENTIFIC BOUNDARIES (HARD CONSTRAINTS):
- MUST NOT VIOLATE: First Law of Thermodynamics (Energy cannot be created or destroyed)
- MUST NOT VIOLATE: Second Law of Thermodynamics (Entropy increases)
- MUST NOT VIOLATE: Speed of light in a vacuum is an absolute limit (c)
- MUST NOT VIOLATE: Conservation of mass and energy
- MUST NOT VIOLATE: Material stress and strain limits
- MUST NOT VIOLATE: Thermodynamic efficiency limits (Carnot limit)

B. VIABILITY SCORE CALCULATION:
The AI proposes an initial viability estimate; the Python Simulation Engine then applies these rules in order:
1. Substance Lock: a description under 50 words scores 0%.
2. Any VERIFIED violation of a hard constraint scores 0%: either the Critic's accusation with a quote checked by Python, or a law checked by arithmetic (heat-to-work efficiency above the Carnot limit of the stated temperatures).
3. Otherwise: AI estimate, minus 15% per logical flaw, 15% per red flag (known pseudo-scientific patterns), 5% per unproven assumption. Accusations Python could not verify are listed and require human review, but do not change the score.
4. A hypothesis is accepted only with a final score of at least 65% and no red flag that contradicts a fundamental law (energy without input, faster-than-light signalling, energy from the vacuum, entropy reversal); those are blocked for human review.
5. Acceptance means automated screening only; accepted hypotheses still require expert review.

C. INVENTOR MODE (speculative exploration):
- The engine explores many mechanism families; every attempt is logged in section 5.
- Each hypothesis must state a numeric, falsifiable PREDICTION and a REFUTATION condition (Falsifiability Lock: otherwise 0%).
- New physical quantities or laws may be postulated, but each must declare SI dimensions, a coupling, and the condition under which it reduces to known physics. Each postulate costs 10%.
- The listed physical laws remain hard constraints: postulates extend them, they never override them.
- An idea stopped by a law is not discarded: a REROUTE attempt must reach the same effect while keeping the law (paying the cost the law demands, relying on a technology that does not exist yet, or changing the

condition the limit depends on). Each missing technology costs 10%.

- Ideas come from the model's own knowledge unless literature is listed in section 1. They are speculative candidates for experiments, not established science.

3. Proposed Hypotheses

Hypothesis 1 [REJECTED] - Viability: 0%

Quantum Vacuum Resonance Trigger

Mechanism family: Electromagnetic cavity excitation

A sub-millimeter cylindrical cavity is constructed from superconducting niobium, with walls polished to atomic-level smoothness and maintained at 0.3 K using adiabatic demagnetization. The cavity is filled with a high-density, low-temperature plasma (electron density ~10^24 m^-3, temperature ~10^0 K) generated by pulsed laser ablation of a tungsten target. A sequence of 100 nanosecond pulses is applied via a Marx generator, delivering a total charge of 100 µC to a central electrode, creating a rapidly rising electric field (up to 10^9 V/m) across the cavity. This field drives intense electromagnetic oscillations at a resonant frequency of 1.2 THz, corresponding to a cavity mode with a spatial confinement of 1 mm in diameter and 1 mm in length. The electromagnetic energy density in this mode reaches ~10^12 W/m^3 within a 1 mm^3 volume due to the high field strength and small volume. The energy is sustained for 10 µs via the cavity's low losses and high Q-factor (Q ~ 10^5), maintained by the superconducting walls and cryogenic environment. The total energy stored in the cavity at peak is ~10^-6 J, consistent with the energy density and volume.

Prediction: A broadband electromagnetic emission spectrum will be measured at the cavity's resonant frequency (1.2 THz) with a power density of at least 10^12 W/m^3 within a 1 mm^3 region, detected via high-sensitivity heterodyne interferometry with a spatial resolution of 0.1 mm and a temporal resolution of 10 ns.

Refutation: If the measured power density in the 1.2 THz band falls below 10^11 W/m^3 over a 1 mm^3 volume during the 10 µs pulse, or if the emission is observed to exceed the local energy density limit defined by the speed of light (c) and the vacuum permittivity (epsilon?), then the mechanism fails to sustain the required energy density. Specifically, if the energy density exceeds 10^12 W/m^3 in any 1 mm^3 volume, the measured power would violate the speed of light limit via the relation u = (1/epsilon?µ?) E^2, where E is the electric field, and such a value would require E > 10^10 V/m, which would produce a radiation pressure exceeding 10^12 Pa - beyond material stress limits of any known dielectric.

Strategy: explore | family: Electromagnetic cavity excitation

Target requirement: A localized energy density of at least 10^12 W/m^3 must be sustained for 10 µs within a 1 mm^3 volume to initiate spacetime distortion.

Audit: AI estimate 35% | logical flaws: 1 | assumptions: 6 | unverified accusations: 1 | human review: yes

Logical flaws: The mechanism asserts a sustained energy density without explaining how electromagnetic energy can be confined and maintained at such levels without radiative losses or instabilities, violating the principle of energy conservation in dynamic EM systems. (quote: 'The energy is sustained for 10 µs via the cavity's low losses and high Q-factor')

Assumptions: The superconducting niobium cavity can maintain atomic-level smoothness and operate stably at 0.3 K under pulsed laser ablation conditions.; The high-density plasma (10^24 m^-3) can be generated and sustained at 10^0 K within the cavity without instabilities or quenching.; The Marx generator can deliver 100 µC of charge in 100 nanosecond pulses without breakdown or energy loss.; The electromagnetic cavity will sustain resonant oscillations at 1.2 THz with a Q-factor of 10^5 over 10 µs without damping or

mode coupling.; The energy density of 10^12 W/m^3 in a 1 mm^3 volume can be achieved and maintained without exceeding material or vacuum limits.; Heterodyne interferometry can detect power densities of at least 10^12 W/m^3 with 0.1 mm spatial and 10 ns temporal resolution.

Unverified accusations (not applied): (law cited in the flaw review, not verified) The target requirement of sustaining 10^12 W/m^3 for 10 µs is contradicted by the second law of thermodynamics, as such a high energy density would produce immense entropy locally, and no known...

Hypothesis 2 [REJECTED] - Viability: 0%

Chiral Spin-Current Collapse

Mechanism family: Spin-Current Confinement

A nanoscale helical lattice of graphene nanoribbons is fabricated in a vacuum chamber, with dimensions of 1 mm in length and 1 mm in width, forming a closed loop with a chiral helical twist. The lattice is maintained at 4 K using liquid helium cooling to minimize thermal noise and enable coherent spin transport. A focused laser pulse of 100 femtoseconds duration and 10 mJ total energy is directed at the lattice to excite spin currents via optical spin injection. This generates a circulating spin current with a density of 10^24 spins per m^3, moving along the helical path at a velocity of 10^5 m/s, driven by the spin-orbit coupling in the graphene structure. The spin current is confined to a 1 mm^3 volume due to the geometric chirality and topological constraints of the lattice, and the spin angular momentum density reaches 10^7 J/m^3. The total energy density in this volume is calculated from the spin current's kinetic energy and magnetic moment, yielding a localized energy density of 10^12 W/m^3 sustained for 10 µs. This energy density arises not from electromagnetic fields but from the kinetic and magnetic energy of coherent spin motion, with no violation of energy conservation during the pulse.

Alternative path (law kept): The earlier idea violated the second law of thermodynamics by implying local entropy decrease; this version avoids it by using spin current energy that is dissipated into the lattice as heat, with entropy increase measured in the surrounding environment, ensuring total entropy rises.

Requires (does not exist yet): Chiral spin lattice with helical topological confinement | SPEC: 10^24 spins/m^3 sustained for 10 µs with no detectable spin leakage

Prediction: A time-resolved magneto-optical Kerr effect will show a peak spin current density of at least 10^12 W/m^3 within a 1 mm^3 region, measured with a 10 ns temporal resolution and 0.1 mm spatial resolution using a polarized probe laser.

Refutation: If the measured spin current density falls below 10^11 W/m^3 over a 1 mm^3 volume during the 10 µs pulse, or if the local entropy change is measured to be less than 10^-6 J/K in the surrounding environment, the idea is refuted.

Strategy: reroute | family: Spin-Current Confinement

Target requirement: A localized energy density of at least 10^12 W/m^3 must be sustained for 10 µs within a 1 mm^3 volume to initiate spacetime distortion.

Audit: AI estimate 30% | logical flaws: 5 | assumptions: 4 | unverified accusations: 0 | human review: no

Logical flaws: The claimed spin angular momentum density of 10^7 J/m^3 contradicts the known energy density of a spin current, as spin angular momentum is not equivalent to energy density and cannot reach such values without violating the spin-statistical and energy-momentum constraints of quantum mechanics. (quote: 'the spin angular momentum density reaches 10^7 J/m^3'); A spin current density of 10^24 spins per m^3 moving at 10^5 m/s implies a kinetic energy density of (1/2)mv^2, which for electrons (mass ~10^-30 kg) yields ~5×10^-16 J/m^3-orders of magnitude below 10^12 W/m^3, making the energy density claim inconsistent with the

actual kinetic energy of spin currents. (quote: 'a circulating spin current with a density of 10^24 spins per m^3, moving along the helical path at a velocity of 10^5 m/s'); The total energy density of 10^12 W/m^3 sustained for 10 µs implies a total energy of 10^-6 J in a 1 mm^3 volume, which is consistent in magnitude, but the claim that this energy arises from 'kinetic and magnetic energy of coherent spin motion' contradicts known physics: spin currents do not carry significant energy densities, and magnetic energy densities in such systems are limited by the saturation of magnetization and the Pauli exclusion principle. (quote: 'the total energy density in this volume is calculated from the spin current's kinetic energy and magnetic moment'); The prediction of a magneto-optical Kerr effect peak at 10^12 W/m^3 is not distinguishable from known spin dynamics in graphene, as such effects are already observed in conventional spintronic systems and do not uniquely validate a localized energy density of 10^12 W/m^3. (quote: 'a peak spin current density of at least 10^12 W/m^3 within a 1 mm^3 region'); The hypothesis asserts that the energy density arises from spin motion without electromagnetic fields, yet spin currents inherently generate magnetic fields, and the energy stored in such fields is governed by the magnetic energy density formula (B^2)/(2µ?), which for any realistic field strength is far below 10^12 W/m^3. (quote: 'This energy density arises not from electromagnetic fields but from the kinetic and magnetic energy of coherent spin motion')

Assumptions: Chiral spin lattice with helical topological confinement can sustain 10^24 spins per m^3 for 10 µs without detectable spin leakage; Optical spin injection via a 100 fs, 10 mJ laser pulse can generate a circulating spin current of 10^24 spins/m^3 at 10^5 m/s in graphene nanoribbons; The spin angular momentum density reaches 10^7 J/m^3 without violating energy conservation or spin dynamics laws; The total energy density of 10^12 W/m^3 is sustained for 10 µs without electromagnetic field contributions or external energy input

Hypothesis 3 [REJECTED] - Viability: 0%

Phase-Modulated Resonant Array

Mechanism family: Electromagnetic Boundary Feedback

A linear array of 1000 identical, thin metallic rods, each 1 meter in length and 1 mm in diameter, is arranged in a straight line with spacing of 1 mm between centers. Each rod is coated with a superconducting oxide layer (e.g., YBCO) and connected to a resonant LC circuit tuned to a frequency of 1.5 GHz, corresponding to a wavelength of 20 cm in free space. A high-voltage pulse (10 kV, 100 ns duration) is applied to each rod sequentially at 100 Hz intervals via a precision Marx generator array, inducing a transient surface current that creates a localized electromagnetic field. The rods are positioned such that their longitudinal axis lies parallel to the 1-meter path, and their mutual proximity generates strong near-field coupling via evanescent waves. The electromagnetic fields from adjacent rods interfere constructively due to a phase shift induced by the time-varying boundary conditions at the rod surfaces, with the phase shift arising from the delay in the current propagation along each rod's length (via the skin effect and surface resistance). The phase shift per rod is calculated using the propagation delay along the rod's length (L = 1 m), with the electromagnetic wavefront advancing at a speed of approximately 2 × 10^8 m/s (a fraction of c), leading to a phase shift of Deltaphi = (2pi / lambda) × (L × f) × (1 - epsilon), where epsilon is a small correction from dispersion and boundary reflection, and the total phase across the 1 m path is accumulated from all rods. The resulting interference pattern is observed in a detection plane 1 m beyond the array, using a high-sensitivity RF antenna array with sub-nanosecond temporal resolution.

Prediction: A measurable interference pattern with a peak-to-peak amplitude of at least 0.5 mV at 1.5 GHz, sustained for a duration of at least 10 milliseconds, observed at a distance of 1 meter from the end of the array, with a spatial coherence length of at least 10 cm.

Refutation: If the detected interference amplitude is less than 0.1 mV at 1.5 GHz, or if the pattern decays to below 0.05 mV within 1 millisecond, or if the spatial coherence length drops below 5 cm, then the phase shift

of at least 0.1 radians across the 1-meter path has not occurred. This would imply that the induced phase modulation is insufficient to produce detectable interference, violating the requirement. Additionally, if the energy input to the system exceeds 100 joules per 1000 pulses (i.e., 100 J/kW·s), then energy conservation is violated, as the output energy would exceed the input energy.

Strategy: explore | family: Electromagnetic Boundary Feedback

Target requirement: A phase shift of at least 0.1 radians must be induced in the electromagnetic field across a 1 m path, resulting in a measurable interference pattern that persists beyond natural decay.

Audit: AI estimate 30% | logical flaws: 3 | assumptions: 1 | unverified accusations: 1 | human review: yes

Logical flaws: The claimed phase shift of 0.1 radians across a 1-meter path at 1.5 GHz contradicts the fundamental physics of electromagnetic wave propagation, as the phase shift from a 1-meter path at 1.5 GHz is inherently fixed and cannot be modulated by surface current delays or skin effect in a way that produces a net phase shift of 0.1 radians without violating the dispersion and boundary conditions of Maxwell's equations. (quote: 'the phase shift per rod is calculated using the propagation delay along the rod's length (L = 1 m), with the electromagnetic wavefront advancing at a speed of approximately 2 × 10^8 m/s (a fraction of c), leading to a phase shift of Deltaphi = (2pi / lambda) × (L × f) × (1 - epsilon)'); The mechanism asserts that time-varying boundary conditions induce a phase shift via current propagation delay, but no physical derivation or known electromagnetic model supports such a phase modulation in a thin metallic rod array under transient high-voltage pulses; this step lacks any explanation of how boundary conditions alter phase in a way that accumulates over rod length. (quote: 'the phase shift arising from the delay in the current propagation along each rod's length (via the skin effect and surface resistance)'); The prediction of a sustained interference pattern with peak-to-peak amplitude of at least 0.5 mV at -1.5 GHz, lasting 10 ms, would be a prediction of known physics in a resonant array, and thus cannot distinguish the hypothesis from conventional electromagnetic interference; it is not a novel or unique prediction. (quote: 'a measurable interference pattern with a peak-to-peak amplitude of at least 0.5 mV at 1.5 GHz, sustained for a duration of at least 10 milliseconds')

Assumptions: The phase shift per rod is calculated using a propagation delay at 2 × 10^8 m/s, which is a fraction of the speed of light, and this leads to a phase shift Deltaphi = (2pi / lambda) × (L × f) × (1 - epsilon) with epsilon being a small correction from dispersion and boundary reflection.

Unverified accusations (not applied): (law cited in the flaw review, not verified) The energy input threshold of 100 joules per 1000 pulses (100 J/kW·s) is inconsistent with the energy required to generate a 1000-pulse sequence at 100 Hz, which would require a total energy input...

4. Conclusive Synthesis

PIPELINE SUMMARY: Evaluated 3 hypotheses. 0 hypotheses passed strict deterministic evaluation.

DEEP ANALYSIS:
Exploration finished (iterations exhausted). 8 attempt(s): 0 accepted, 8 rejected, 0 discarded as duplicates, 0 invalid format, 0 assessment failures.
Accepted per strategy: explore 0/3, recombine 0/1, postulate 0/1, reroute 0/3.
Average time per attempt: 0.6 min.
Requirements derived from the goal (model-derived, not verified):
R1: A localized energy density of at least 10^12 W/m^3 must be sustained for 10 μs within a 1 mm^3 volume to initiate spacetime distortion. -> 6 attempt(s), 0 accepted, best 0%
R2: A net transfer of at least 5 × 10^15 J must occur between two points separated by 100 m in less than 1 ms, violating local energy conservation unless mediated by a non-local mechanism. -> 0 attempt(s), 0

accepted, best n/a
R3: A phase shift of at least 0.1 radians must be induced in the electromagnetic field across a 1 m path, resulting in a measurable interference pattern that persists beyond natural decay. -> 2 attempt(s), 0 accepted, best 0%
A candidate for one requirement does not achieve the goal: the goal needs ALL requirements at once.
No candidate passed the screening. That is a valid result: it means the explored mechanisms did not survive, not that the goal is impossible.
Closest attempt 1: Quantum Vacuum Resonance Trigger [explore] 0%.
Closest attempt 2: Chiral Spin-Current Collapse [reroute] 0%.
Closest attempt 3: Phase-Modulated Resonant Array [explore] 0%.
What would have to be invented first (declared by the candidates, not verified):
- Chiral spin lattice with helical topological confinement | SPEC: 10^24 spins/m^3 sustained for 10 μs with no detectable spin leakage
- Ultra-high-current superconducting pulse coil | SPEC: 100 kA peak current, 10 μs pulse duration, 99.9% efficiency in energy transfer from input power to plasma kinetic energy
- Electron-velocity confinement array | SPEC: 10^10 m/s electron velocity in vacuum
Most common barriers: grounded logical flaws (5); deterministic red flags (2); too many unproven assumptions (1).

5. Exploration Map

Requirements derived from the goal (model-derived, not verified):
R1: A localized energy density of at least 10^12 W/m^3 must be sustained for 10 μs within a 1 mm^3 volume to initiate spacetime distortion.
R2: A net transfer of at least 5 × 10^15 J must occur between two points separated by 100 m in less than 1 ms, violating local energy conservation unless mediated by a non-local mechanism.
R3: A phase shift of at least 0.1 radians must be induced in the electromagnetic field across a 1 m path, resulting in a measurable interference pattern that persists beyond natural decay.

#1 EXPLORE [R1] | Electromagnetic cavity excitation | rejected 0%
barrier: AI estimate 35% -> final 0% after penalties (logical flaws: 1, assumptions: 6, unverified accusations: 1, red flags: 0, numeric errors: 0); below the 65% threshold. 1 Critic accusation(s) could not be verified, so the AI estimate may reflect unproven claims; human review recommended. Flaw: The...
#2 REROUTE [R1] | improves #1 | Spin-Current Confinement | rejected 0%
barrier: AI estimate 30% -> final 0% after penalties (logical flaws: 5, assumptions: 4, unverified accusations: 0, red flags: 0, numeric errors: 0, missing technologies: 1); below the 65% threshold. Flaw: The claimed spin angular momentum density of 10^7 J/m^3 contradicts the known energy density of a spin...
#3 EXPLORE [R3] | Electromagnetic Boundary Feedback | rejected 0%
barrier: AI estimate 30% -> final 0% after penalties (logical flaws: 3, assumptions: 1, unverified accusations: 1, red flags: 0, numeric errors: 0); below the 65% threshold. 1 Critic accusation(s) could not be verified, so the AI estimate may reflect unproven claims; human review recommended. Flaw: The...
#4 POSTULATE [R1] | Topological Energy Coupling | rejected 0%
barrier: AI estimate 15% -> final 0% after penalties (logical flaws: 3, assumptions: 5, unverified accusations: 2, red

flags: 1, numeric errors: 0, postulates: 1); below the 65% threshold. 2 Critic accusation(s) could not be verified, so the AI estimate may reflect unproven claims; human review...

#5 REROUTE [R1] | improves #4 | Magnetic Shear Induced Discontinuity | rejected 0%

barrier: AI estimate 15% -> final 0% after penalties (logical flaws: 2, assumptions: 1, unverified accusations: 2, red flags: 2, numeric errors: 0, missing technologies: 1); below the 65% threshold. 2 Critic accusation(s) could not be verified, so the AI estimate may reflect unproven claims; human review...

#6 RECOMBINE [R3] | seeds: topological protection + synthetic opal + strong 10 T magnetic field | Topological Opal Resonance | rejected 0%

barrier: AI estimate 15% -> final 0% after penalties (logical flaws: 2, assumptions: 6, unverified accusations: 1, red flags: 0, numeric errors: 0); below the 65% threshold. 1 Critic accusation(s) could not be verified, so the AI estimate may reflect unproven claims; human review recommended. Flaw: The...

#7 EXPLORE [R1] | Phase-locked microcrystalline arrays | rejected 0%

barrier: AI estimate 15% -> final 0% after penalties (logical flaws: 0, assumptions: 5, unverified accusations: 1, red flags: 0, numeric errors: 0); below the 65% threshold. 1 Critic accusation(s) could not be verified, so the AI estimate may reflect unproven claims; human review recommended.

#8 REROUTE [R1] | improves #7 | High-velocity plasma sheath excitation | rejected 0%

barrier: AI estimate 15% -> final 0% after penalties (logical flaws: 2, assumptions: 4, unverified accusations: 2, red flags: 0, numeric errors: 0, missing technologies: 1); below the 65% threshold. 2 Critic accusation(s) could not be verified, so the AI estimate may reflect unproven claims; human review...